



Inspiring Excellence

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Course: CSE 489

Name: Farjana Sadia Prome

ID: 22301007

Email:

farjana.sadia.prome@g.bracu.ac.bd

Project Title: Urban Houseplant Care & Watering Schedule App**Project Features:**

1. Core Dashboard & NavigationCentral Hub (dashboard_screen.dart): A premium-styled dashboard that provides a quick overview of the user's "gardening status."

Stats Summary: Displays the total number of plants and tasks completed today.

Quick Actions: Large, interactive cards for navigating to the Marketplace, AI Expert, Care Guide, and Chat system.

- Upcoming Tasks: A dynamic list of care tasks (Watering, Misting) filtered by urgency, with clear "Overdue" indicators.
- Recent Plants: A horizontal carousel of the user's most recently interacted-with plants.
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2. Plant Management System

- Personal Inventory (plant_list_screen.dart): Allows users to view their entire collection of plants.
- Plant Entry (add_edit_plant_screen.dart): A form-based interface to add new plants or update existing ones, including names, species, and watering frequencies.
- Detailed View (plant_detail_screen.dart): Shows comprehensive data for a single plant, including its care history and growth status.

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3. AI-Powered Features (services/ai_service.dart)

- AI Expert Chat (ai_assistant_screen.dart): A rule-based conversational agent (designed for offline/local use) that provides expert advice on watering, light requirements, and troubleshooting plant health issues (e.g., yellowing leaves).
- Plant Identification: An interface that uses the device camera to capture a plant image and identifies it (currently a mock implementation for offline use).

4. Care Scheduling & Reminders

- Care Scheduler (services/care_scheduler.dart): Logic that calculates when a plant needs its next watering or misting based on user-defined frequencies.
- Notification System (services/notification_service.dart): Manages local notifications to remind users when it's time to care for their plants, ensuring no plant is forgotten.
- Task Tracking: Ability to "mark as done" directly from the dashboard, which automatically updates the plant's history.

5. Comprehensive Care Guide

- Offline Database (care_guide_service.dart): A built-in library of ~25 plants (loaded via assets/data/care_guide.json) providing instant access to care instructions, difficulty levels, and sunlight needs.
- Wikipedia Integration: An online search feature that fetches real-time summaries and images from Wikipedia for plants not found in the local database.
- Categorization: Plants are organized into categories (e.g., Succulents, Tropicals) for easier browsing.

6. Marketplace & Community

- Marketplace Dashboard (marketplace_screen.dart): A dedicated space for users to browse plants listed for sale by other community members.

- User-to-User Chat (chat_screen.dart, chat_list_screen.dart): A full messaging system allowing buyers and sellers to communicate directly within the app.

7. Growth Tracking

- Growth Diary (growth_diary_screen.dart): A dedicated screen where users can record height, health status, and notes over time to visualize their plant's progress.

8. Technical Foundation

- State Management: Uses the Provider package for reactive UI updates across the app (Auth, Plants, Themes).
- Local Persistence: Utilizes Hive (a lightweight NoSQL database) for storing user plants, care history, and chat messages locally on the device.
- User Authentication: Includes full registration and login flows (login_screen.dart, register_screen.dart).

Feature: Plant Growth & Progress Diary

This feature lets users track their plant's journey from a tiny seedling to a full-grown houseplant, all stored locally on their phone.

Key Offline Capabilities:

1. Visual Timeline: Take periodic photos of each plant. The app displays them in a "Time-lapse" style gallery so users can see how much the plant has grown.
2. Growth Notes: Add offline notes for specific milestones (e.g., "Nov 12: First new leaf!" or "Dec 5: Repotted into a larger ceramic pot").
3. Local Storage: All images and notes are stored in your existing Hive database and the device's local file system, so it works perfectly without an internet connection.

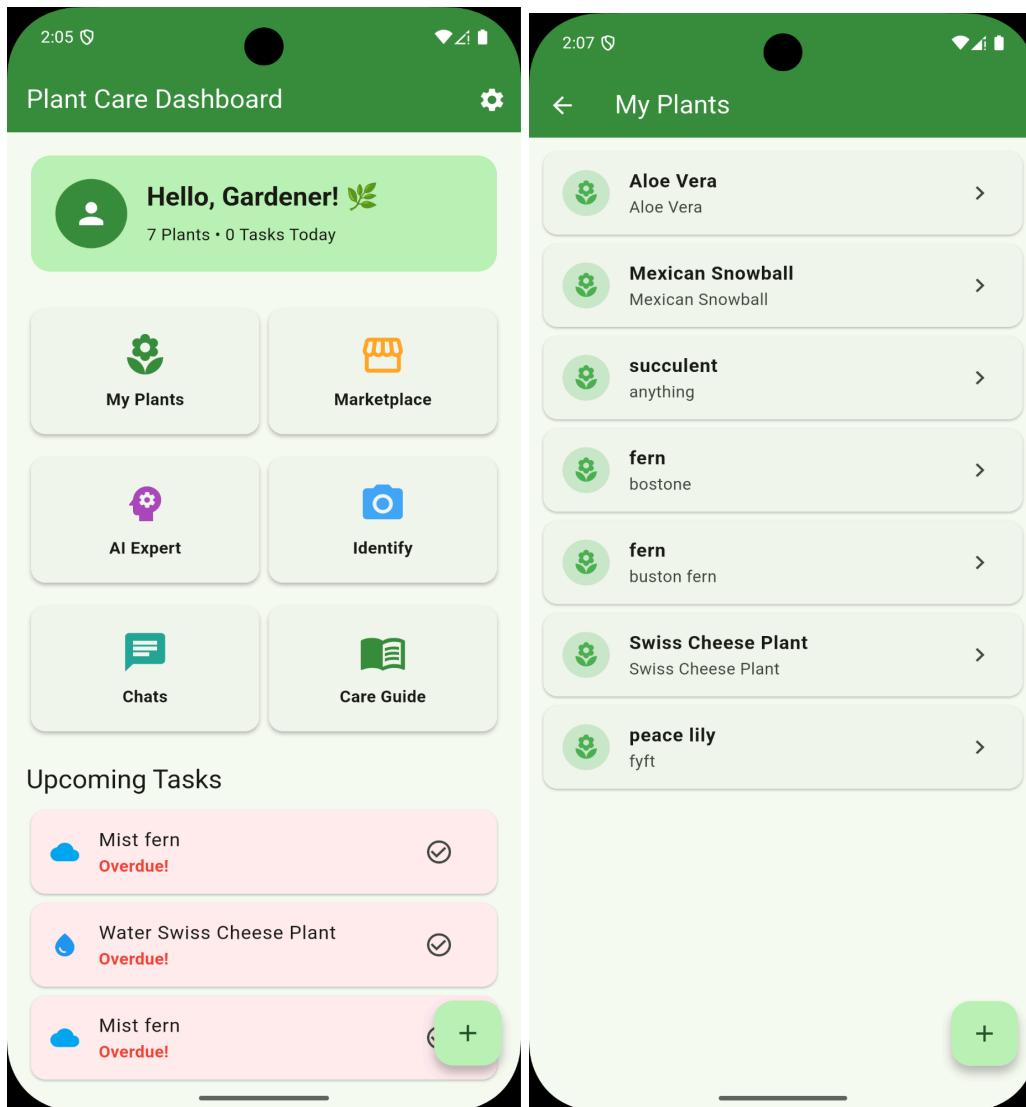
Why this is a great addition:

- Emotional Connection: It turns the app from a simple "utility" (reminders) into a personal "memory book" for their plants.
- Premium Look: A scrolling horizontal timeline of photos looks very modern and professional.
- High Utility: It helps users notice if a plant is struggling over time or if its growth has slowed down.

How I can implement it:

1. Update the Database: We add a List<GrowthEntry> to your Plant model (or create a new Hive box for growth logs).
2. Add a Gallery Tab: Add a new section in the PlantDetailScreen to view the photo history.
3. Capture UI: A simple button to "Add Growth Photo" which saves the image path locally.

Screenshots:



Screenshot 2: Add/Edit Plant Screen

2:06

← Add Plant



Care Intervals (Days) - Enter 0 to disable

 Water Every X Days

 Mist Every X Days

 Fertilize Every X Days

SAVE PLANT

2:31

←



Aloe Vera

Aloe Barbadensis Miller

Succulent

Easy

 Toxic to Pets

Care Instructions

 **Watering**
Every 2-3 weeks

 **Details**
Water deeply but infrequently. Allow soil to dry out completely between waterings. Reduce watering in winter. Plant in terracotta pot for best drainage.

Light & Environment

 **Sunlight**
Bright, direct to indirect light. 6+ hours of sunlight.

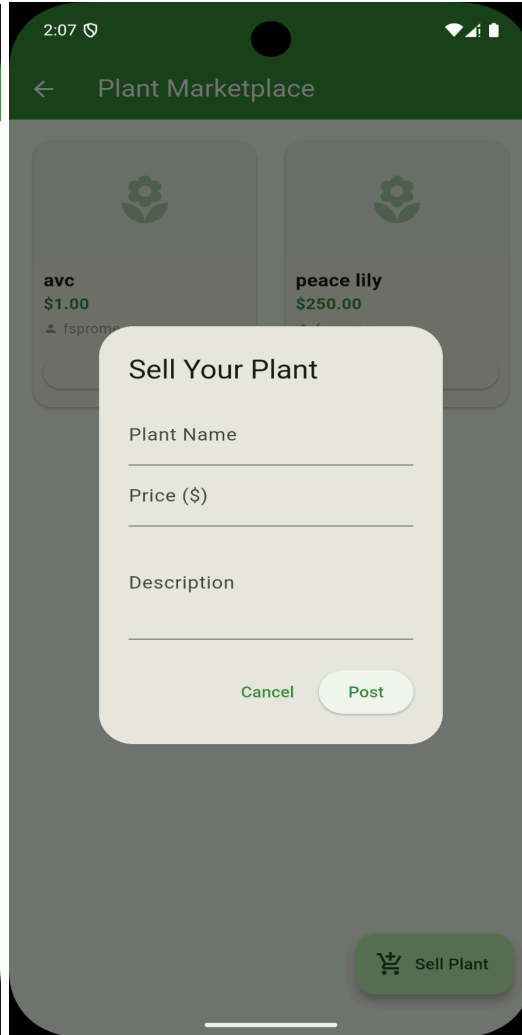
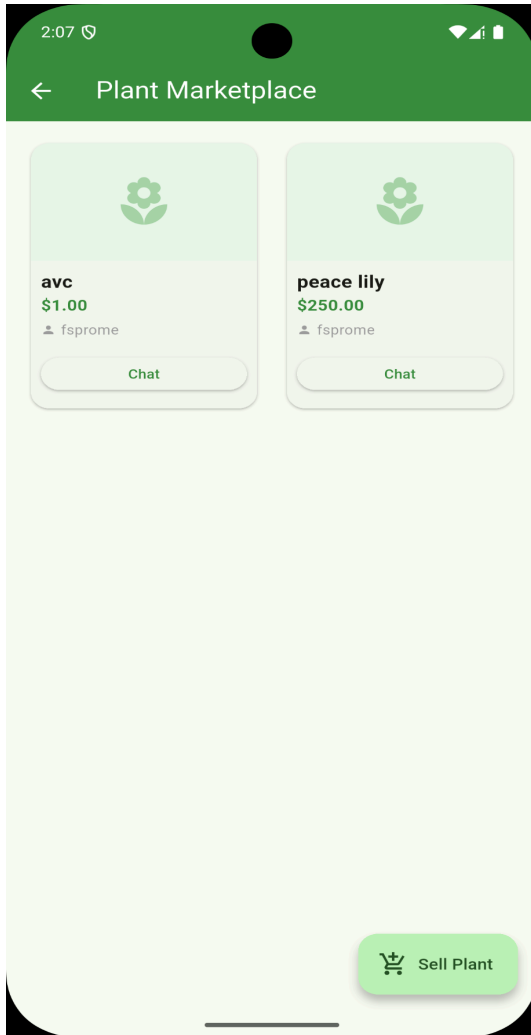
 **Humidity**
Low. Prefers dry conditions.

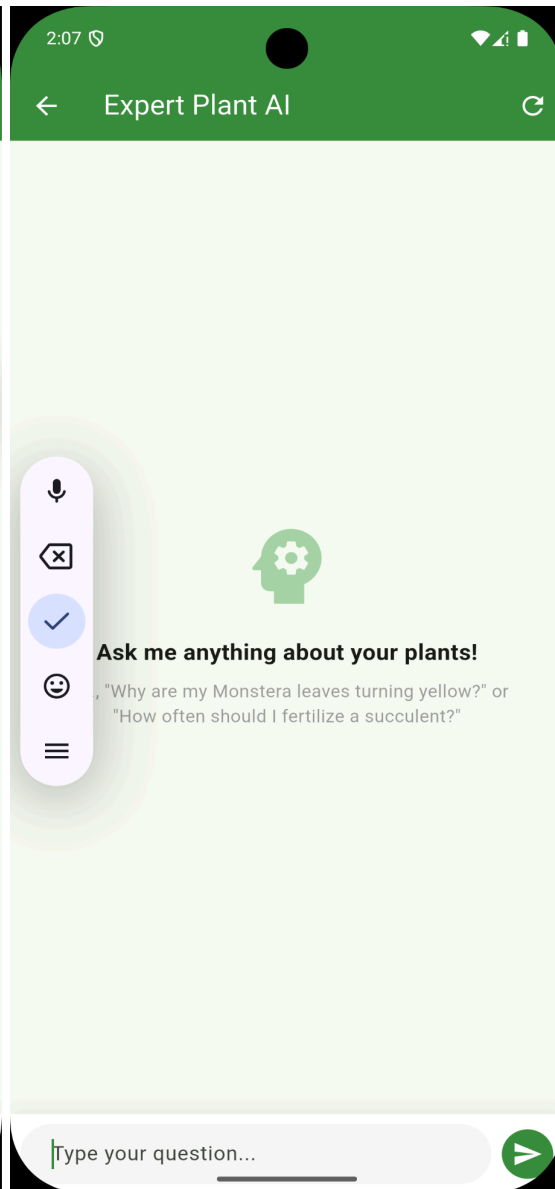
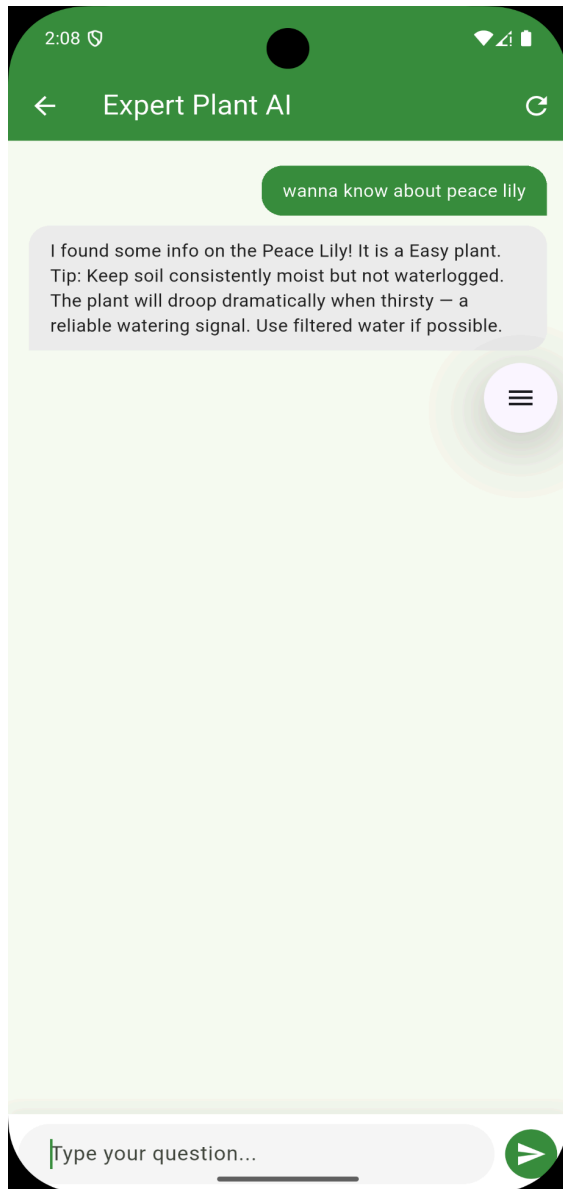
 **Soil**
Sandy, well-draining cactus mix.

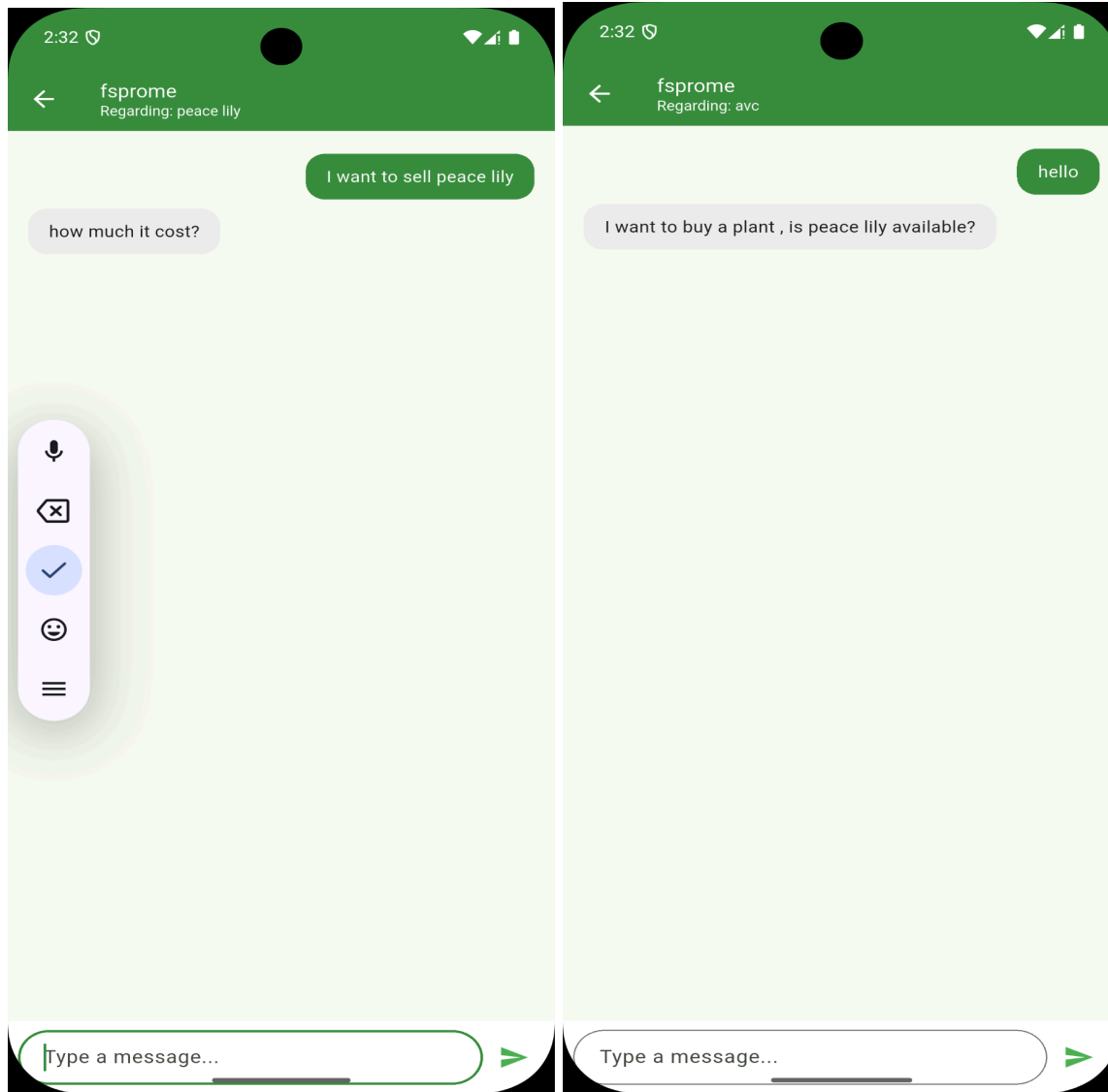
 **Safety & Toxicity**

 Toxic to cats and dogs. Gel is safe for humans

+ Add to My Plants







Online Resources used:

a) Reference:

- [Flutter Official Documentation](#)
- [Hive Documentation for Local Storage](#)
- YouTube link 1: [Flutter Local Notifications Tutorial](#)
- Wikipedia REST API (for live plant data fetching)

b) Stackoverflow or GitHub links:

- flutter_local_notifications GitHub Repository:
https://github.com/MaikuB/flutter_local_notifications
- Hive Flutter setup and usage guides from StackOverflow for dynamic scheduling logic.

- Reference UI Design from Freelancer:
<https://www.freelancer.com/projects/graphic-design/urban-gardening-app-case-study>

Future Enhancements:

The following enhancements can be added to the current system to improve the overall functionality and user experience:

1. Light/Dark Sensor Integration: Utilizing the phone's native ambient light sensor to automatically inform the user if the current spot has enough light for a specific plant.
2. Data Backup & Restore: Allowing users to export their local Hive database to a JSON file or Google Drive so they don't lose their plant data when changing devices.
3. Advanced Plant Recognition: Integrating a local, on-device machine learning model (like TensorFlow Lite) to identify a plant's species automatically using the camera instead of manually searching.
4. Weather API Integration: Automatically skipping scheduled watering notifications if the local weather forecast predicts high humidity or rain for outdoor/balcony plants.